INDUSTRY PROJECT REPORT

FAKE NEWS DETECTION USING SOCIAL MEDIA DATA

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Abstract

The widespread dissemination of fake news on social media platforms poses a serious threat to information credibility and public trust. Manual verification of news content is inefficient at scale, creating a need for automated detection systems. This project proposes an automated fake news detection framework using Natural Language Processing (NLP) and supervised machine learning techniques. News articles are pre-processed and transformed into numerical feature representations using Term Frequency–Inverse Document Frequency (TF–IDF). A Logistic Regression classifier is trained to distinguish between real and fake news based on linguistic and stylistic patterns present in the text. The model is evaluated using standard performance metrics including accuracy, precision, recall, and F1-score, achieving high classification performance within the dataset scope. A lightweight interactive frontend is implemented using Streamlit to enable real-time user interaction and prediction. Experimental results demonstrate the effectiveness of the proposed approach while also highlighting limitations related to dataset bias and stylistic variation across news sources.

Introduction

The rapid growth of social media platforms has transformed the way information is created, shared, and consumed. While these platforms enable fast dissemination of news, they have also facilitated the widespread circulation of fake news, defined as false or misleading informationpresented in the form of legitimate news. Fake news can influence public opinion, distort facts, a

Traditional methods of verifying news content rely heavily on manual fact-checking by experts or organizations. Although effective on a limited scale, such approaches are time-consuming, resource-intensive, and impractical for handling the vast volume of information generated daily on social media. As a result, there is a growing need for automated systems capable of identifying and filtering fake news efficiently.

Recent advancements in Natural Language Processing (NLP) and machine learning have enabled the development of automated text analysis systems that can learn patterns from large datasets. By analysing linguistic features and writing styles, these systems can distinguish between real and fake news with considerable accuracy. This project explores the application of NLP techniques and supervised machine learning to build an automated fake news detection system that operates effectively within the constraints of social media data.

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Problem Statement

The The rapid dissemination of fake news across social media platforms has significantly challenged the credibility of online information. Due to the high volume and speed at which content is generated and shared, manual verification of news articles is inefficient, timeconsuming, and impractical at scale. This creates a critical need for an automated system capable of analyzing textual content and accurately distinguishing between real and fake news. Such a system must operate efficiently on large datasets and provide reliable predictions based on linguistic and stylistic features present in news articles.

Objectives

- To detect fake news using Natural Language Processing techniques.
- To build a supervised machine learning classifier.
- To evaluate model performance using standard metrics.
- To develop an interactive frontend for real-time prediction.

Dataset Description

The data preprocessing steps included converting text to lowercase, removing URLs, punctuation, and extra whitespace. The title and text fields were combined into a single content column to enhance contextual understanding.

Feature Extraction

TF–IDF (Term Frequency–Inverse Document Frequency) was used to convert textual data into numerical features. This method captures the importance of words while reducing the impact of common terms.

Model Development

A Logistic Regression classifier was used due to its effectiveness with high-dimensional and sparse text data. The dataset was split into training and testing sets using an 80–20 ratio.

Model Evaluation

The model achieved an accuracy of approximately 99%. Evaluation metrics included precision,

System Architecture

User Input → Text Cleaning → TF–IDF Vectorization → Logistic Regression Model → Prediction

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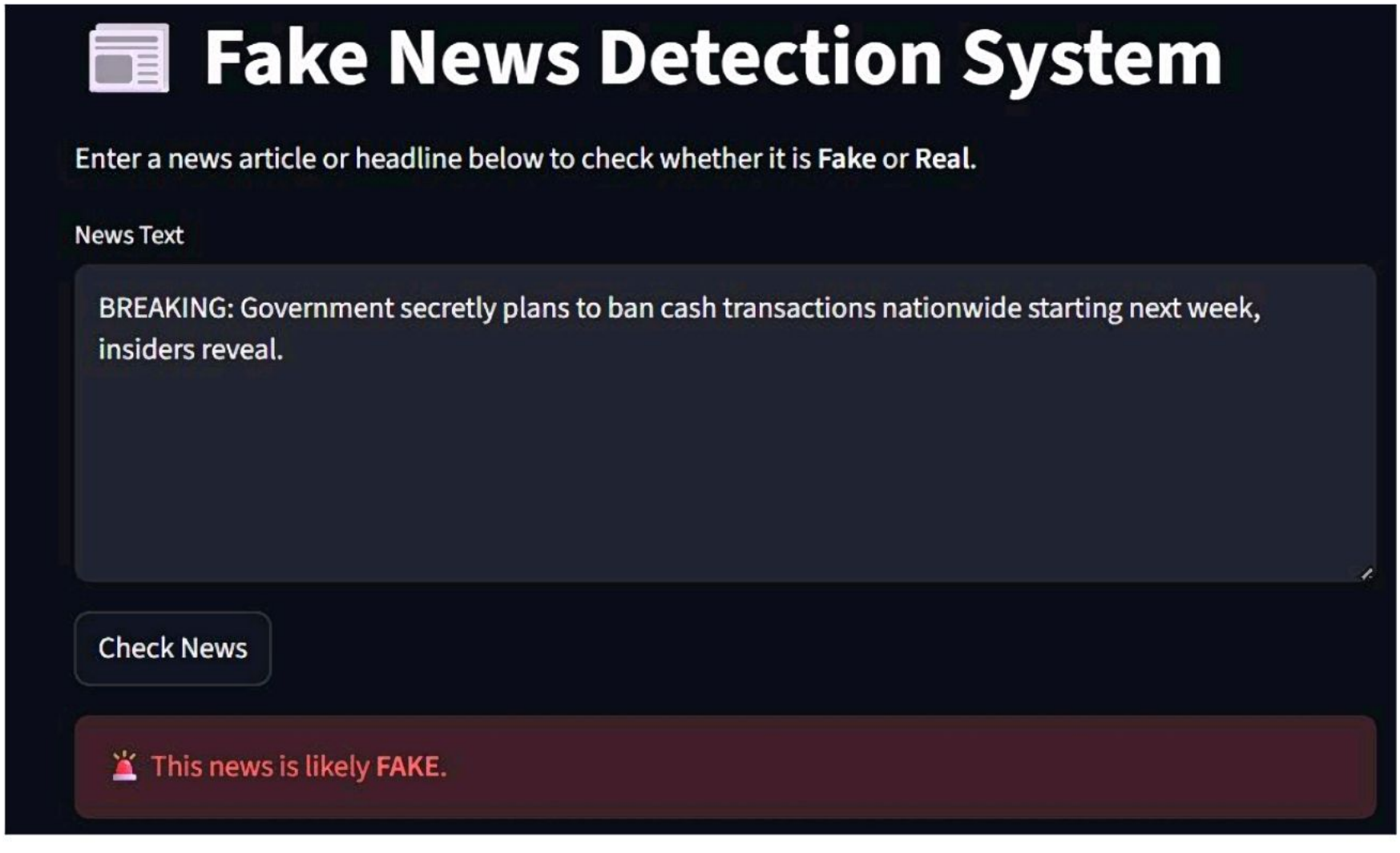
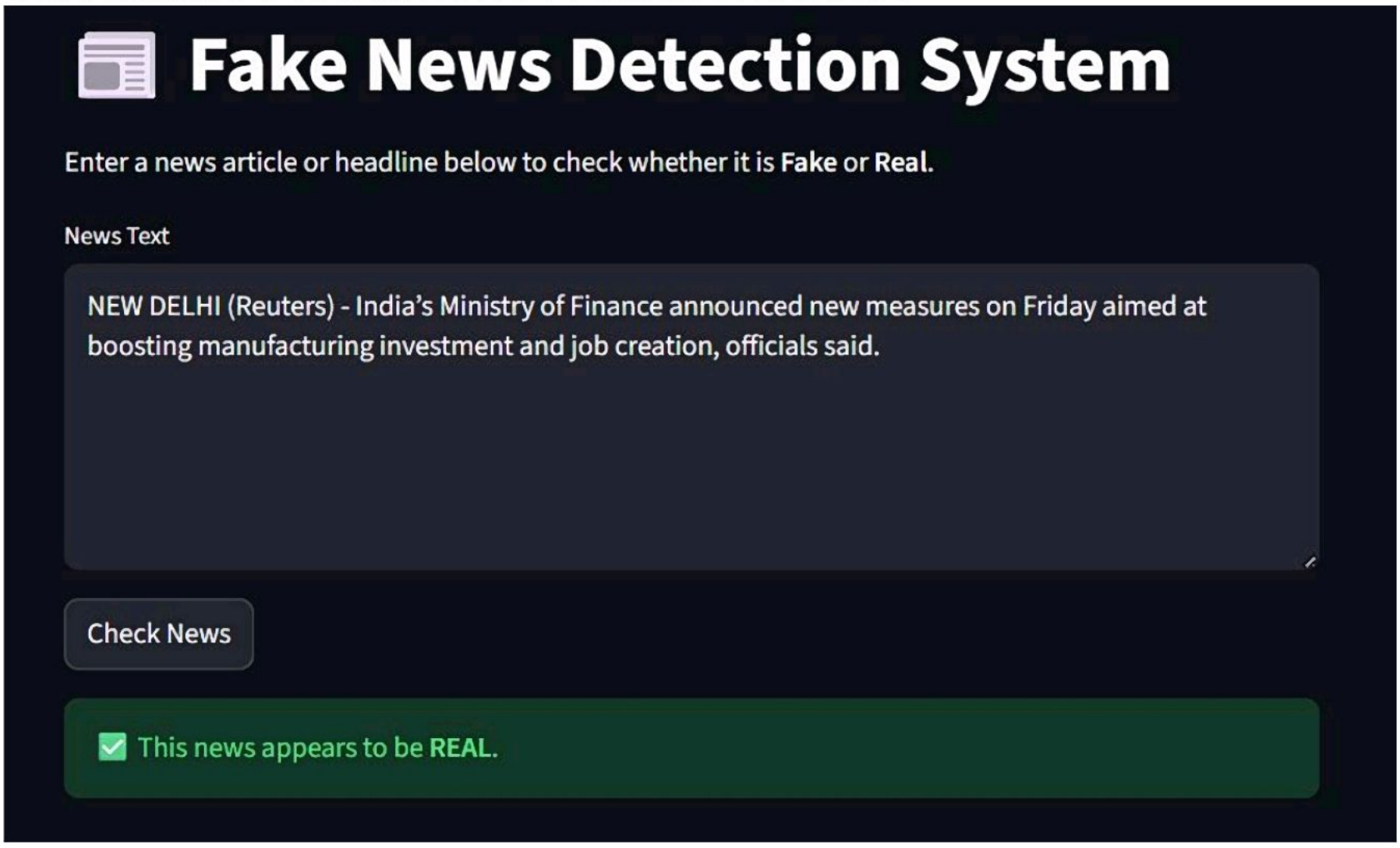
FRONTEND IMPLEMENTATION

Screenshot 1:Streamlit Home Screen

screenshot 2:Fake News Prediction Result



Screenshot 3 : Real News Prediction Result



A Streamlit-based frontend was developed to allow users to input news text and receive p
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Testing

Testing was conducted using predefined test scenarios, test cases, and a test design document to validate system behavior

Conclusion

The project successfully demonstrates an automated fake news detection system using NLP and □m

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References

- Kaggle Fake and Real News Dataset
- Natural Language Toolkit (NLTK) documentation
- Scikit-learn Documentation
- Streamlit Documentation3

Test Case Documentation

TS-01: Fake News Classification

- Description: Validate that the system correctly identifies fake news articles based on
  textual patterns.
- Component: ML Model + NLP Pipeline
- Expected Outcome: Fake news articles are classified as Fake

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TS-02: Real News Classification

- Description: Validate that the system correctly identifies real news articles written in
  formal journalistic style.
- Component: ML Model + NLP Pipeline
- Expected Outcome: Real news articles are classified as Real.

TS-03: User Input Handling

- Description: Validate that the frontend accepts user input and processes it without
  errors.
- Component: Streamlit Frontend
- Expected Outcome: System accepts input text and returns a prediction.

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TS-04: End-to-End System Integration

- Description: Validate the complete flow from user input to final prediction output.
- •Component: Frontend + Backend Model
- Expected Outcome: Input text is successfully classified in real time.

Test Cases -Fake News Detection System

Test Case ID	Test description	Input text	Expected Output	Actual Output	status
TC-01	Verify classification of clearly fake news	BREAKING: Scientists confirm earth will stops rotating tomorrow	Fake	Fake	Pass
TC-02	Verify classification of formal real news text	WASHINGTON (Reuters):The U.S Federal Reserve signaled that interests rates could remain high as inflation moderates	Real	Real	Pass
TC-03	Verify handling of generic news style input	The government announced a new economic policy to reduce inflation	Real	Fake	Fail
TC-04	Verify system response to empty input	No input	warning message	warning message	Pass

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Test Design -Fake News Detection System

Test Design ID	Test scenario ID	Test Description	Test Case ID (s)	Expected Result	Status
TD-01	**TS-01**	Validation of fake news classification	**TC-01**	Fake news is correctly classified	**Pass**
TD-02	**TS-02**	Validation of real news classification	**TC-02**	Real news is correctly classified	**Pass**
TD-03	**TS-03**	Validation of user input handling	**TC-03**	System handles empty input gracefully	**Pass**
TD-04	**TS-04**	End -to -end system validation	**TC-04**	System classifies news in real time	Pass with limitations